



University of Vavuniya

Second Examination in Information Technology - 2022

^{Risk}
Second Semester - January/February 2024

IT2143 Visual Programming (Theory)

First Question is compulsory and answer any One from rest of the questions

Time Allowed: One hour

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- PUFF A
1. (a) State five key characteristics of *Rapid Application Development (RAD)*. [20%]
 - (b) Define the concept of '*Scope Creep*' in the context of (*RAD*). ^{time cut} [15%]
 - * (c) Discriminate between System programmers and Application programmers. [10%]
 - (d) Explain the significance of AI integration in Visual programming and provide an example of how AI can enhance the development process. [20%]
 - (e) Briefly describe four benefits of *Prototyping* in Application Development. ^{time ↓} ^{Front User Sketch} ^{Updating and user involve work} [20%]
 - (f) What does *Common Language Runtime (CLR)* signify in the .NET framework?. [15%]
 2. (a) Describe how *DLLs* facilitate code re-usability and modularity in .NET applications. [20%]
 - * (b) Explain the process of User Interface design with a flow diagram. [30%]
 - (c) Differentiate between the Reactive and Pro-active risk management strategies in software development with the aid of suitable examples. [20%]

[This question is continued on the next page]

User friendly
User audience
User Diversity
Consistency
Minimal Super
Reliable.

- ✓ (d) Briefly describe the role of the three primary components of the Model-view-controller architecture in software development, and list three web development frameworks that implement the MVC architecture. [30%]
3. (a) Explain the steps involved in the process of compilation in .NET. [25%]
- PCL
CLR (b) Define what is meant by *Silver-bullet Syndrome* in your own words, and discuss its impact on technology within the realm of software development. [25%]
- (c) Provide suitable examples each of the following User Interface design principles.
- i. Minimal Surprise
 - ii. Consistency
 - iii. Recoverability
 - iv. User Guidance [20%]
- (d) Assume, you are assigned as a software architect to design a sophisticated web-based application for a financial institution, Explain the potential impact of poor coding practices in this project. [30%]